



Clean Out Your Ears!

Filtering out Noise from Speech Samples



There is sound all around us!

- Often, too much background noise can make it very difficult to understand what a speaker is saying
- It can drown the speaker out with just general noise such as construction, cars, wildlife, or even other people talking
- So, what if we could remove this noise?

Dataset used

Valentini Noisy dataset

(<https://www.Kaggle.com/datasets/muhmagdy/valentini-noisy>)

- Used the noisy and clean test sets from the dataset
- 824 samples each, paired between the noisy and clean
- 2 speakers, 1 male, 1 female
- Additionally contains 4 other sets, 2 each noisy and clean, for training a ML model – total of 70.9k samples

Algorithm

- Started by resampling each file to a 16kHz sample rate
- Passed each resampled file through 4th order butterworth filter:
 - Low edge: 300Hz
 - High edge: 3400Hz
- Processed each filtered file using a spectrum subtraction:
 - Converted the audio signal to frequency domain using an FFT
 - Estimated the noise spectrum by using the first $\frac{1}{4}$ second, before the speaker starts
 - Subtract this noise spectrum from the signal
 - Reconstruct the audio signal through inverse FFT
 - Normalize the output to a range of [-1, 1]

Results

